PBBI clause extraction — fine-tuned v0.1.3 vs gpt-4.1-mini

Test set: 17 Perjanjian Baku contracts never used in training · **Date:** 2026-10-17 · **Both models at temperature 0**

Summary

The fine-tuned model **v0.1.3** (the current `stable` release) finds **2.3 times as many** of Legal's clauses as **gpt-4.1-mini**, and more of what it returns is right: **F1 0.716 vs 0.391**. It has the higher F1 on 13 of the 17 contracts and ties on 2. gpt-4.1-mini wins on 2 (BRI UE Chip-Based and OCBC Internet Banking).

	v0.1.3 (fine-tuned Qwen-SEA-LION 27B + LoRA)	gpt-4.1-mini
Recall: Legal's clauses found	**0.658** (1,365 / 2,074)	0.283 (587 / 2,074)
Precision: returned clauses that are right	**0.784** (1,365 / 1,741)	0.632 (587 / 929)
F1	**0.716**	0.391
Clauses returned ÷ labelled	**0.84**	0.45
Valid JSON / finished on its own	17/17 / 17/17	17/17 / 17/17
Higher F1 on	**13** contracts (2 ties)	2
Output tokens, all 17	230,894	186,125
Time for the longest contract	15.8 min (H200, 8 at a time)	2.8 min
Cost of a 17-contract run	≈ $2 (H200 pod, ~25 min incl. model load)	≈ $0.45 (API)

Temperature. Both runs here use temperature 0, so each model gives its single most likely answer and a re-run gives the same output. gpt-4.1-mini does **better at its default temperature, 1.0**: F1 **0.470** (recall 0.349, precision 0.720), as measured in the v0.1.2 report. Even at its best setting it is well behind v0.1.3.

The clause-by-clause workbook

`pbbi-extraction-clauses-v0.1.3-vs-gpt-4.1-mini.xlsx`, in this folder, shows **what each model generated**, next to Legal's label, for every clause of the 17 contracts:

Sheet	Contents
Ringkasan	The scores above, and how to read the other sheets
Per kontrak	Per contract: returned, matched, recall, precision, F1, minutes, for both models
Perbandingan klausa	One row per clause Legal labelled (No. Pasal, Topik, Bunyi), with v0.1.3's clause and gpt-4.1-mini's clause beside it. Green means match; red means missed, with the closest clause shown and its similarity. After each contract, the clauses each model returned that match none of Legal's are listed in orange
v0.1.3 (fine-tuned) (mentah) / **gpt-4.1-mini (mentah)**	Each model's output exactly as generated, in order

The workbook quotes the contracts word for word. It is gitignored (`docs/pbbi/reports/*.xlsx`) and must be shared the way the source contracts are, not through git. This page and the csv hold scores and bank names only.

Results per contract

Contract	Penyelenggara	Labelled	v0.1.3 found / returned	v0.1.3 F1	mini found / returned	mini F1
JATENG Kartu Debit	Utama / BPD	15	15 / 15	1.000	15 / 15	1.000

Contract	Penyelenggara	Labelled	v0.1.3 found / returned	v0.1.3 F1	mini found / returned	mini F1
MUAMALAT Internet Banking	Utama / Swasta Nasional	46	46 / 46	**1.000**	32 / 35	0.790
BCA Kartu Debit	Utama / Swasta Nasional	87	79 / 80	**0.946**	50 / 57	0.694
SINARMAS UE Server-Based	Utama / Swasta Nasional	49	44 / 49	**0.898**	4 / 14	0.127
MANDIRI Mobile Banking	Utama / BUMN	98	88 / 102	**0.880**	53 / 68	0.639
NAGARI Mobile Banking	BPD	44	37 / 44	**0.841**	27 / 32	0.711
BRI UE Chip-Based	Utama / BUMN	27	25 / 33	0.833	25 / 31	**0.862**
SULTRA Kartu Debit	BPD	18	15 / 18	0.833	15 / 18	0.833
OY Payment Gateway	Swasta Nasional	89	67 / 86	**0.766**	20 / 32	0.331
SEABANK Mobile Banking	Utama / Swasta Nasional	358	247 / 290	**0.762**	42 / 66	0.198
SHOPEEPAY UE Server-Based	Utama / LSB	211	130 / 147	**0.726**	57 / 84	0.386
DBS Mobile Banking	Utama / Multinational	262	168 / 207	**0.716**	95 / 162	0.448
SMBC Kartu Kredit	Utama / Swasta Nasional	172	127 / 197	**0.688**	45 / 107	0.323
OCBC Kartu Kredit	Utama / Multinational	233	131 / 154	**0.677**	19 / 44	0.137
BJB Mobile Banking	Utama / BPD	110	69 / 116	**0.611**	4 / 19	0.062
BOC Mobile & Internet Banking	Utama / Multinational	79	25 / 41	**0.417**	18 / 33	0.321
OCBC Internet Banking	Utama / Multinational	176	52 / 116	0.356	66 / 112	**0.458**
All 17		**2,074**	**1,365 / 1,741**	**0.716**	**587 / 929**	**0.391**

By product

Product	Contracts	v0.1.3 F1	gpt-4.1-mini F1
Kartu Debit	3	**0.936**	0.762
Uang Elektronik Chip-Based	1	0.833	**0.862**
Payment Gateway	1	**0.766**	0.331
Uang Elektronik Server-Based	2	**0.763**	0.341
Mobile Banking	5	**0.747**	0.363
Kartu Kredit	2	**0.683**	0.230
Internet Banking	2	0.510	**0.531**
Mobile Banking & Internet Banking	1	**0.417**	0.321

By penyelenggara

Penyelenggara	Contracts	v0.1.3 F1	gpt-4.1-mini F1
Utama / BUMN	2	**0.869**	0.696
BPD	2	**0.839**	0.750
Utama / Swasta Nasional	5	**0.790**	0.349
Swasta Nasional	1	**0.766**	0.331

Penyelenggara	Contracts	v0.1.3 F1	gpt-4.1-mini F1
Utama / LSB	1	**0.726**	0.386
Utama / BPD	2	**0.656**	0.239
Utama / Multinational	4	**0.593**	0.360

By contract length

Labelled clauses	Contracts	v0.1.3 recall	gpt-4.1-mini recall
under 100	10	**0.799**	0.469
100–199	3	**0.541**	0.251
200 or more	4	**0.635**	0.200

What the results show

1. **gpt-4.1-mini returns less than half of the clauses.**
2. It returns 929 clauses where Legal labelled 2,074 (0.45×), and finds 587 of them.
3. On four contracts it finds almost nothing: BJB (4 of 110), SINARMAS (4 of 49), OCBC Kartu Kredit (19 of 233), SEABANK (42 of 358).
4. Every reply ended normally (`stop`), well under its 32,768-token limit, so it is not being cut off. It merges sub-clauses and stops early. The *Perbandingan klausa* sheet shows where.
5. **v0.1.3 follows Legal's level of detail.** It returns 0.84× as many clauses as Legal, and is near-perfect on short contracts: recall 0.80 on contracts with fewer than 100 clauses.
6. **The gap is largest where it matters most.**
7. On contracts with 200 or more clauses, v0.1.3 finds 0.64 of Legal's clauses and gpt-4.1-mini finds 0.20.
8. A clause the extraction misses is never reviewed for compliance, so recall is the number to watch.
9. **gpt-4.1-mini wins on 2 contracts.**
10. **OCBC Internet Banking** (0.458 vs 0.356) is v0.1.3's known weak point: Legal groups its definitions into ranges, and v0.1.3 gives one clause per term (see the v0.1.3 report).
11. **BRI UE Chip-Based** (0.862 vs 0.833) is a 27-clause contract where the two are within one clause of each other.
12. **gpt-4.1-mini is worse at temperature 0 than at 1.0** (F1 0.391 vs 0.470). With its single most likely answer it stops even earlier. Sampling sometimes carries it further through the document.

How the test was run

- **Models.**
- **v0.1.3:** `scalejade/pbbi-clause-extraction-lora` tag `v0.1.3` (`stable`, commit `63773a48`) on the base `scalejade/qwen-sea-lion-v4.5-27b-pbbi` at `8d32a4b9`. It ran on vLLM 0.30, the production engine, on one H200.
- **gpt-4.1-mini:** the OpenAI API.
- **Request.** Both models got the app's exact extraction call: `extraction_system.md` byte for byte, then `"DOCUMENT:\n"` + the app's PDF text, with `response_format: json_object` and `temperature: 0`.
- Output limit: v0.1.3 got the app's 65,536 tokens. gpt-4.1-mini got 32,768, its API maximum. None of its replies came near that limit.
- **Scoring.**
- Every reply goes through the app's own `parse_reply` and `finalize`.
- A clause counts as found when it shares at least 80% of its words with a labelled clause, matched one to one, closest pairs first.
- Legal's labels scored against themselves get 100%.
- The workbook is built by `preparation/compare_runs.py` with the same matching, and its totals equal these.

Caveats

- **17 test contracts.** Several products and penyelenggara types have only one or two, so group-level differences are indicative.
- **gpt-4.1-mini at its default temperature** scores higher (F1 0.470) than in this temperature-0 comparison. Either way, v0.1.3 leads by more than 0.24 F1.
- **The match threshold** (≥ 0.8 word overlap) is strict about how clauses are split. A merged or differently split clause counts as a miss even when its words are present.

Files

- `pbbi-extraction-eval-v0.1.3-vs-gpt-4.1-mini.{md,csv,pdf}`: this report and the per-contract numbers (scores and bank names only). The csv also has gpt-4.1-mini's temperature-1.0 F1 per contract.
- `pbbi-extraction-clauses-v0.1.3-vs-gpt-4.1-mini.xlsx`: every clause each model generated, beside Legal's. **Gitignored: it quotes the contracts.**
- Raw runs (gitignored): `preparation/data/scores/test-v0.1.3-t0/`, `test-gpt-4.1-mini-t0/`, `test-gpt-4.1-mini/` (temperature 1.0). To rebuild the workbook:

```
python preparation/compare_runs.py \
  preparation/data/scores/test-v0.1.3-t0 "v0.1.3 (fine-tuned)" \
  preparation/data/scores/test-gpt-4.1-mini-t0 "gpt-4.1-mini" \
  --out docs/pbbi/reports/pbbi-extraction-clauses-v0.1.3-vs-gpt-4.1-mini.xlsx
```